

Andrew Strachan

Birmingham, Alabama | 228-224-7445 | strachan@uab.edu

EDUCATION

Mississippi College, Clinton, MS

B.S. ACS Biochemistry | Biology Medical Sciences | Music (minor) | Graduated with Honors: May 2016

University of Montevallo, Montevallo, AL

Provisional Certificate in a Teaching Field - Business, Marketing, and Finance | 2024 to 2025

University of Mississippi School of Medicine, Jackson, MS

Doctor of Medicine (incomplete) | 2016 to 2020

University of Alabama Birmingham, Birmingham, AL

Masters of Cybersecurity | Expected Graduation: Dec 2027

TECHNICAL SKILLS

- **AI/ML:** On-Device Execution (MLX, Ollama), RAG, Vector Search, Agentic Workflows, Prompt Eng, Python, OpenAI/Google AI APIs
- **Dev & Full Stack:** Swift/SwiftUI (macOS/iOS), TypeScript, React, Node.js, C, Java, RESTful API, Git/GitHub
- **Cybersecurity:** Wireshark, Digital Forensics, OSINT, SIEM, Nmap, Zero-Trust, Linux SysAdmin
- **Cloud & Data:** GCP, AWS, Firebase, SQL, SwiftData, ChromaDB, FAISS, NoSQL Schema Design

PROFESSIONAL SUMMARY

- Current M.S. in Cybersecurity student at UAB with a dual background in biomedical sciences, four years of medical school training, and hands-on full stack and AI development experience. Proven ability to bridge complex medical and research-facing workflows with practical software solutions, secure data handling, and technical operations in regulated environments. Experienced in building AI-assisted applications, developing websites and digital workflows, leading training efforts, and translating technical requirements into usable systems and documentation.

MaqkrTutor - Recent Project Highlight - <https://github.com/astrachan163/MaqkrTutor-public.git>

- Personal macOS AI Application; SwiftUI, SwiftData, Ollama, Gemma 4, RAG, SQLite vector retrieval, with Markdown rendering.
 - Built a local macOS AI study application for grounded interactions over PDFs, Markdown, and code files using on-device model workflows and retrieval-augmented generation.
 - Implemented document ingestion, direct-context and vector-based retrieval, focused-document selection, target-language controls, and model-aware chat behavior for predictable summarize, translate, explain, and practice-question workflows.
 - Designed the interface with a course-workspace UI to support contextualized learning and practice-question generation.
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PROFESSIONAL EXPERIENCE

University of Alabama Birmingham

Birmingham, AL

Python Coding Summer Camp Teaching Assistant

June 2026

- Instructed middle school students in fundamental Python programming concepts, data structures, and algorithmic logic through hands-on STEM projects, guiding students to program BBC micro:bit microcontrollers to control sensors, LED displays, and external hardware.
- Taught practical hardware-software integration by mentoring students through writing custom Python scripts to program and execute flight trajectories for aerial drones.
- Fostered a collaborative, problem-solving environment that increased student engagement and technical confidence in computer science.

Shades Valley High School

Birmingham, AL

CTE Teacher: Business, Marketing, and Finance

Aug 2024 - June 2025

- Taught business finance and customer service & sales in our National Academy of Finance, and served as coach for the E-Sports team; organized fundraisers for competitions and equipment.
- Maintained TAPE compliance in all areas of Career and Technical Education instruction, and served as adviser for the state's largest Future Business Leaders of America chapter.

Corner High School

Dora, AL

CTE Teacher: Business, Marketing, and Finance

Sept 2023 - June 2024

- Established and advised a new DECA chapter, fostering student leadership, professional development, and community engagement. Coached and mentored students in DECA competitions, culminating in first and second-place achievements at the 2024 Alabama DECA Career Development Conference.
- Maintained TAPE compliance in all areas of Career and Technical Education instruction.

MidSouth Extracts LLC.

Tupelo, MS

Operational Director

Jan 2023 - May 2023

- Built a licensed medical processing facility, procured state and federal licensing, and translated technical processes into a cohesive regulatory compliant ecosystem, authoring a complete library of SOPs. Executed a comprehensive task and process analysis within a cGMP-regulated manufacturing environment by embedding with and interviewing diverse SME teams (engineers, Ph.D. chemists, technicians).

SelectQuote Insurance Services

Clinton, MS

Sales Development Specialist

Apr 2021 - Nov 2022

- Developed and implemented a comprehensive training program for new sales representatives, collaborated with cross-functional teams to identify knowledge gaps in the sales process and created targeted training modules, leading to a reduction in customer complaints and an improvement in overall customer satisfaction.

University of Mississippi School of Medicine

Jackson, MS

Human Anatomy Prosector

May 2017 - Sep 2017

- Developed and delivered educational presentations to a class of 100 students, effectively showcasing complex anatomical concepts and enhancing understanding. Assisted with and demonstrated dissection procedures and developed educational presentations to create compelling coursework, providing hands-on learning experiences for students in the medical school program.

- Operational consulting including expertise in FDA, ISO, and employee training processes; building new startups along with their foundational processes for websites, advertising, sales, licensure, and technology.
- Facilitating work-study and data-annotation internships to support large-scale biomedical research. Managing entry-level data sanitation, image labeling, and text formatting pipelines under strict supervision to support the development of Large Language Models (LLMs) allotting students real-world experience in clinical decision support. Focused on STEM diversity by directing work-based learning in basic vulnerability scanning, log monitoring, and compliance checklist verification to maintain zero-trust architectures.

Mississippi College
Human Gross Anatomy Teacher Assistant

Clinton, MS
Aug 2015 - May 2016

- Evaluated student work objectively and provided constructive feedback to enhance their understanding of human anatomy concepts, leading to improvement in student performance over the course of two semesters.
- Supervised and guided students through lab experiments, ensuring accurate execution and adherence to safety protocols while maintaining a secure learning environment.

First Presbyterian Church of Gulfport
Assistant Youth Leader

Gulfport, MS
Aug 2011 - Present

- Provided mentorship to a diverse group of young individuals facing academic challenges, supporting them through personalized guidance and resources. Organized and facilitated team-building activities for a group of 30 students, fostering effective communication and teamwork skills, leading to an improvement in collaborative project outcomes of mentored youth through various challenges and overcoming academic obstacles.

VOLUNTEER EXPERIENCE

Omnimed Health Initiative
Medical Intern

Uganda, Africa
Jul 2017 - Aug 2017

- Utilized strong cross-cultural communication skills to effectively convey complex scientific concepts to diverse groups during health workshops, fostering understanding and engagement among participants from different cultural backgrounds.
- Collaborated with local leaders and healthcare professionals to identify and address critical healthcare needs in rural communities, resulting in the establishment of two mobile clinics that served over 1,000 patients annually. Led team efforts to develop and implement comprehensive health education workshops, reaching over 500 community members and increasing health literacy rates.

Travel the World
Classroom Travel Correspondent

Pokhara, Nepal
Jun 2014 - Aug 2014

Shared my experiences as a Benjamin A. Gilman Scholarship fellow with New York elementary students through video and live classroom Skype sessions, including:

- B.G. Hospital & Research Center Medical Intern - Utilized practical medical knowledge to assist in clinical duties, resulting in improved patient outcomes and a decrease in medication errors.
- Machapuchre Development Organization Volunteer Community Educator - Traveled to underdeveloped villages in the Himalayan to teach farmers modern organic farming methods, increase crop yield, and delivered hands-on instruction in modern organic farming techniques to over 50 farmers in remote villages.

Harrison County Children's Shelter
Volunteer Youth Mentor

Gulfport, MS
May 2011 - Jan 2018

- Facilitated a supportive learning environment, encouraged students to explore scientific interests, mentored at-risk youths, provided guidance on developing long-term goals and life skills.
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